POLITECNICO DI TORINO · Master Degree in Data Science

Official Examination Session 2

Date: June 19, 2026 | **Time Allotted:** 1 Hour 30 Minutes | **Total Points:** 16.00 Points

Session Scope
Summer Examination Session

§1 Exam Session 2 — June 19, 2026

Session Overview: This examination comprises 7 questions covering actionable counterfactuals, gradient targets in backpropagation, permutation feature importance and its mathematical limitations, faithfulness versus plausibility, label-free concept-based architectures (ProtoPNet vs. SENN vs. VLMs), physical-world adversarial threats, and an operational XAI system design for financial tabular fraud detection (total: 16.00 points).

Question 1: Actionability in Counterfactual Explanations — MCQ — 1.00 Point

Which of the following counterfactuals is **least** actionable?

a. Decrease your age by 10 years.

b. Reduce the requested loan amount.

c. Increase savings.

d. Reduce current debt.

✓ Executive Solution Summary

The correct answer is **a**. Chronological age is a strictly **immutable feature** governed by irreversible temporal physical laws. A counterfactual recommendation requiring an individual to decrease their age by 10 years cannot be enacted under any real-world human intervention, rendering it completely non-actionable.

➤ Systematic Analysis of Multiple Choice Options

(a) [CORRECT] Decrease your age by 10 years.
Actionability requires that the feature modifications specified by the counterfactual can be practically executed by the end user. Age is strictly non-actionable and monotonically non-decreasing. Asking an applicant to be 10 years younger provides zero recourse.

(b) [INCORRECT] Reduce the requested loan amount.
Actionable: A loan applicant can immediately modify the application form to request a smaller principal loan amount.

(c) [INCORRECT] Increase savings.
Actionable (over time): An applicant can accumulate financial capital and deposit additional funds into savings accounts.

(d) [INCORRECT] Reduce current debt.

Actionable: An applicant can pay off credit cards or settle outstanding liabilities to improve their debt-to-income ratio.

✍ **Comprehensive Technical Walkthrough**

In counterfactual recourse research (Wachter et al., Ustun et al.), input features $\mathcal{X}$ are formally partitioned into three distinct operational categories:

$$\mathcal{X} = \mathcal{X}_{\text{immutable}} \cup \mathcal{X}_{\text{actionable}} \cup \mathcal{X}_{\text{conditionally-mutable}} \tag{1}$$

Category	Mathematical Constraint	Domain Examples	Recourse Validity
Immutable	$\Delta x_j = 0$ strictly	Age, Race, Biological Sex, Birth Country	Never admissible in actionable recourse
Actionable	$\Delta x_j \in [\delta_{\min}, \delta_{\max}]$	Loan Amount, Credit Cards, Spending	Direct and immediate user control
Conditionally Mutable	Directional: $\Delta x_j \geq 0$	Education Level, Years at Current Job	Monotonic; requires temporal delay

When generating a counterfactual $x^* = \arg\min_{x'} d(x, x') + \lambda \mathcal{L}(f(x'), y^*)$, an optimization algorithm must enforce an infinite penalty or hard constraint on immutable features:

$$x_j^* = x_j \quad \forall j \in \mathcal{X}_{\text{immutable}} \tag{2}$$

Recommending a decrease in age violates both causal directionality and physical feasibility.

★ **Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)**

Students occasionally confuse *effort* with *actionability*. Increasing savings or paying off debt requires significant effort and time, but it remains fundamentally actionable. Decreasing chronological age is physically impossible regardless of human effort.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 9: "Counterfactual Explanations", Section: "Properties of Counterfactual Explanations: Actionability and Feasibility" (pp. 130–135).

¶ **Alternative Perspectives & Methodological Variants**

Actionable Recourse Frameworks (e.g., Ustun et al., 2019) formulate recourse as an integer programming problem over a user-specific Actionability Graph, mathematically guaranteeing that the solver never explores perturbations along immutable coordinates.

Question 2: Gradient Computation in Explainability vs. Training MCQ 1.00 Point

In gradient-based explainability, gradients are computed with respect to:

a. The training set size

b. The number of classes only

c. The optimizer learning rate

d. The model input features

✓ Executive Solution Summary

The correct answer is **d**. In gradient-based explainability, gradients are computed with respect to the **model input features** ($\nabla_x S_c(x)$). This measures the local sensitivity of the output class score to infinitesimal perturbations of each input dimension, serving as a first-order feature attribution map.

➤ Systematic Analysis of Multiple Choice Options

(d) [CORRECT] The model input features

Gradient-based explainability computes $\nabla_x S_c(x) = \frac{\partial S_c(x)}{\partial x}$. Backpropagating the target class score back to the input features x quantifies how much each pixel, token, or tabular feature influences the prediction.

(a) [INCORRECT] The training set size

Training set size is a fixed scalar dataset property, not a differentiable input variable.

(b) [INCORRECT] The number of classes only

The number of classes is an architectural discrete integer determining the output layer dimension.

(c) [INCORRECT] The optimizer learning rate

The learning rate η is an optimization hyperparameter controlling parameter update step sizes during training, completely irrelevant to feature attribution at inference time.

✍ Comprehensive Technical Walkthrough

A fundamental conceptual distinction exists between standard backpropagation during model training and backpropagation during post-hoc explainability:

Dimension	Model Training (Optimization)	Model Explainability (Attribution)
Target Variable	Training Loss $\mathcal{L}(f_\theta(x), y)$	Target Class Score / Logit $S_c(x)$
Differentiation Target	Model Weights θ ($\nabla_\theta \mathcal{L}$)	Input Features x ($\nabla_x S_c(x)$)
Objective	Update parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$	Attribute importance: $A_i(x) = \frac{\partial S_c(x)}{\partial x_i}$
Mathematical Basis	Empirical Risk Minimization	First-order Taylor series approximation of $f(x)$

By taking the first-order Taylor expansion around the input x:

$$S_c(x + \Delta x) \approx S_c(x) + \nabla_x S_c(x)^T \Delta x \tag{3}$$

The vector of partial derivatives $\nabla_x S_c(x)$ directly represents the linear sensitivity of the model's output to perturbations along each input feature coordinate.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

Do not confuse the gradient of the loss with respect to parameters ($\nabla_\theta \mathcal{L}$), used to train neural network weights, with the gradient of the prediction with respect to the input ($\nabla_x S_c(x)$), used to explain predictions.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 5: "Gradient-based Explainability Methods", Section: "Vanilla Gradient: Gradients: Training vs. Explainability" (pp. 66–69).

¶ Alternative Perspectives & Methodological Variants

Methods like **Grad-CAM** compute gradients with respect to intermediate convolutional feature activation maps ($\frac{\partial Y^c}{\partial A^k}$) rather than raw input pixels, capturing higher-level spatial semantic patterns while filtering high-frequency noise.

Question 3: Permutation Feature Importance: Mechanics, Interpretation, and Scope

Open Inquiry | 2.00 Points

Explain permutation feature importance. Describe the main idea behind the method and how the resulting explanations should be interpreted. Specify whether the explanation is local or global.

✓ Executive Solution Summary

Permutation Feature Importance (PFI) is a **post-hoc, global, model-agnostic** explainability technique. The core idea is to measure the increase in model prediction error after randomly shuffling (permuting) the values of a specific feature across the dataset, thereby breaking the relationship between that feature and the target outcome.
A larger drop in performance indicates that the model heavily relies on that feature for its predictions. PFI is strictly **global** because it computes a single scalar importance score per feature aggregated over the entire dataset. Critical limitations include unrealistic off-manifold sampling for correlated features and high stochastic variance.

✍ Comprehensive Technical Walkthrough

1. Categorization and High-Level Concept

- **Timing / Access:** Strictly **post-hoc** (applied to an already-trained model f) and **model-agnostic** (requires only black-box querying capabilities).
- **Scope:** Strictly **global** (summarizes feature importance across the entire population $\mathcal{D}$).
- **Intuition:** If a feature is critical to the model's predictive power, destroying its information content will cause a substantial increase in prediction error. Conversely, shuffling an uninformative or noise feature will leave the model's performance largely unchanged.

2. Step-by-Step Algorithmic Procedure

Let $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$ be a dataset (preferably a validation or test set) and $\mathcal{L}(y, f(X))$ be a chosen performance loss metric (e.g., Mean Squared Error, Cross-Entropy, or $1 - \text{AUC}$):

Step 1. Compute Baseline Loss: Evaluate the model on the original, uncorrupted dataset:

$$L_{\text{orig}} = \mathcal{L}(y, f(X)) \tag{4}$$

Step 2. Permute Feature Column: For each feature $j \in \{1, 2, \ldots, p\}$:

a. Construct a modified feature matrix $\tilde{X}^{(j)}$ by applying a random permutation π to the values of column j, while holding all other $p - 1$ columns identical:

$$\tilde{X}^{(j)} = [x_{\cdot,1}, \ldots, \pi\left(x_{\cdot,j}\right), \ldots, x_{\cdot,p}] \tag{5}$$

This preserves the marginal distribution of feature j while completely breaking its statistical association with the target y and its joint dependencies with other features.

b. Compute the permuted dataset error:

$$L_{\text{perm}}^{(j)} = \mathcal{L}\left(y, f\left(\tilde{X}^{(j)}\right)\right) \tag{6}$$

c. Compute the raw importance score as either the **difference** or the **ratio**:

$$I_{\text{diff}}(j) = L_{\text{perm}}^{(j)} - L_{\text{orig}} \quad \text{or} \quad I_{\text{ratio}}(j) = \frac{L_{\text{perm}}^{(j)}}{L_{\text{orig}}} \tag{7}$$

Step 3. Variance Reduction: To mitigate the stochasticity of a single random permutation, repeat the shuffling K times and compute the mean and standard deviation:

$$\bar{I}(j) = \frac{1}{K}\sum_{k=1}^{K} I^{(k)}(j), \quad \sigma_I(j) = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}\left(I^{(k)}(j) - \bar{I}(j)\right)^2} \tag{8}$$

3. Interpretation of Results

- $\bar{I}(j) \gg 0$: High positive importance. The model's accuracy deteriorates severely without this feature.
- $\bar{I}(j) \approx 0$: Negligible importance. Shuffling this feature has no measurable effect on prediction error.
- $\bar{I}(j) < 0$: Negative importance. Shuffling the feature actually improves performance, indicating that the feature acts as noise or causes overfitting.

4. Critical Limitations of Permutation Feature Importance

Lim 1. The Correlation Problem and Off-Manifold Sampling: PFI implicitly assumes that features are independent. When features are strongly correlated (e.g., *engine displacement* and *number of cylinders*), shuffling column j independently creates synthetic, physically impossible combinations (e.g., an 8-cylinder engine with a 0.8L displacement). The model is forced to evaluate data points far outside its training manifold, producing unpredictable extrapolation errors that artificially distort the importance score.

Lim 2. Underestimation of Collinear Features: If two redundant, highly correlated features provide the same predictive information, shuffling one allows the model to rely entirely on the intact copy. Consequently, both features may receive misleadingly low importance scores.

Lim 3. Stochastic Instability: Because permutation is random, results exhibit high variance unless averaged over a sufficiently large number of repetitions (K), increasing computational cost.

Lim 4. Loss Function Dependency: Feature importance scores depend entirely on the selected metric $\mathcal{L}$.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

This question had two severe penalties in official exam grading:

1. **Classifying PFI as In-Modeling (-0.30 points):** PFI is strictly **post-hoc** and **model-agnostic**. Stating that PFI is an "in-modeling" method is a fatal theoretical error.
2. **Omitting Limitations (-0.30 points):** Students who described the shuffling mechanism but failed to mention the off-manifold sampling dilemma, correlated features, and stochastic variance were heavily penalized.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 5: "Post-Modeling Explainability", Section: "Permutation Feature Importance (PFI): Computation Steps, Advantages, and Limitations" (pp. 34–37).

¶ Alternative Perspectives & Methodological Variants

Conditional Permutation Importance (CPI) (Strobl et al.) preserves the correlation structure by permuting feature j only within partitions conditioned on correlated covariates, preventing the creation of off-manifold synthetic points.

Question 4: Faithfulness vs. Plausibility in Explanation Evaluation

Open Inquiry | 1.50 Points

Explain the difference between faithfulness and plausibility in the evaluation of explanations. Can an explanation be plausible but not faithful? Motivate your answer.

✓ Executive Solution Summary

Faithfulness (fidelity) measures how accurately an explanation reflects the true internal reasoning mechanism and decision logic of the predictive model. **Plausibility** (convincingness) measures how intuitive, believable, and aligned with human domain knowledge an explanation appears to a human expert.

Yes, an explanation can be highly plausible but completely unfaithful. This dangerous disconnect occurs when an explanation presents reasoning that humans expect to see, while the model internally made its prediction based on spurious correlations, dataset shortcuts, or imaging artifacts. Canonical examples include the TED framework, edge-detecting saliency maps, and adversarial explanation scaffolding.

✍ **Comprehensive Technical Walkthrough**

1. Formal Definitions

Criterion	Faithfulness (Fidelity / Truthfulness)	Plausibility (Convincingness / Human Alignment)
Evaluation Target	Ground truth of the **model's inner computation**	Ground truth of **human domain knowledge**
Core Question	"Does this explanation accurately describe *how the model actually reasoned*?"	"Does this explanation make sense and sound *convincing to a human expert*?"
Validation Method	Objective perturbation tests, feature ablation, sanity checks, axiomatic verification	Human subject studies, clinical expert reviews, user trust surveys
Target Audience	AI auditors, safety engineers, system debuggers	End-users, domain specialists, non-technical clients

Formally, let f be the model, x an input instance, and $E(f, x)$ the generated explanation.

- **Faithfulness** evaluates the functional fidelity:

$$\text{Fidelity}\left(E(f,x), f\right) \approx 1 \iff \forall \Delta x,\ E(f,x) \text{ correctly predicts } f(x+\Delta x) - f(x) \tag{9}$$

- **Plausibility** evaluates alignment with human prior knowledge $\mathcal{K}_{\text{human}}$:

$$\text{Plausibility}\left(E(f,x), \mathcal{K}_{\text{human}}\right) = \text{Similarity}\left(E(f,x), e_{\text{expert}}\right) \tag{10}$$

2. Why Explanations Can Be Plausible But Unfaithful (Concrete Demonstrations)

An explanation can easily score perfectly on plausibility while having zero faithfulness. This represents the most dangerous failure mode in XAI, creating unwarranted human overtrust:

Case 1. **Teaching Explanations for Decisions (TED Framework):** In the TED paradigm, neural networks are trained with a multi-task loss to output both a prediction and a human-authored explanation string. Suppose a clinical chest X-ray model detects pneumonia based purely on a spurious metallic hospital token in the upper corner. Because TED was trained on human medical justifications, it may generate the explanation: *"Diagnosis justified by diffuse bilateral infiltrates in lower pulmonary lobes."*

To an attending physician, this explanation is **100% plausible** because it matches textbook medical pathology. However, it is **0% faithful**: the model's actual computational trigger was the metallic hospital marker, which the explanation completely concealed.

Case 2. **Saliency Map Sanity Checks (Adebayo et al.):** Many visual saliency maps (such as Guided Backpropagation) act as generic edge detectors. When visualizing an image of a Siberian Husky, the heatmap sharply outlines the husky's face and ears. Humans rate this explanation as highly plausible. However, cascading weight randomization tests show that even when all convolutional weights are randomized, the visualization remains identical! If the actual model relied on the presence of snow in the background, the heatmap was totally unfaithful.

Case 3. Adversarial Scaffolding Attacks (Slack et al., 2020): An adversary can construct a racially discriminatory loan classification model and wrap it in a prediction scaffold. When queried on typical user inputs, the model executes biased logic. However, when the scaffold detects off-manifold perturbation points generated by LIME or SHAP, it temporarily switches to a benign egalitarian model. The resulting explanation exhibits low racial feature attribution (**plausible and legally compliant**), but is **utterly unfaithful** to the true deployed logic.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

Prof. Eliana Pastor docked −0.40 points from candidates who provided imprecise or tautological definitions of faithfulness. Common errors include:

- Confusing faithfulness with *accuracy* (faithfulness is NOT whether the model predicts the correct label; an unfaithful explanation can accompany an accurate prediction, and a faithful explanation can describe an inaccurate prediction).
- Defining faithfulness as "the model's capacity to understand the task". Faithfulness is a property of the **explanation**, measuring whether the explanation truthfully reflects the model's inner mechanics.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 6: "Evaluation of Explanations", Section: "Plausibility vs. Faithfulness" (pp. 82–86) & Chapter 4: "In-Modeling Explainability: TED" (pp. 28–30).

¶ Alternative Perspectives & Methodological Variants

Quantitative metrics to rigorously measure faithfulness independent of human subjectivity include **Pixel Flipping / Feature Erasure AUC curves** (measuring output drop as attributed features are removed) and the **Infidelity Metric** (Sen et al., 2020):

$$\text{INFD}(E, f, x) = \mathbb{E}_{I \sim \mu_I} \left[\left(I^T E(f, x) - (f(x) - f(x - I)) \right)^2 \right] \tag{11}$$

Question 5: Explainable-by-Design Concept Models Without Concept Labels

Open Inquiry | 3.00 Points

You want to create an explainable-by-design model using concepts as explanations. However, you do not have concept labels on the training dataset. Describe at least a few approaches that can be employed in this scenario.

✓ Executive Solution Summary

When building **explainable-by-design** models without ground-truth concept annotations, four primary methodological paradigms can be utilized:

1. **Prototype-Based Learning (ProtoPNet / Case-Based In-Modeling):** Learns representative latent prototype vectors that are projected onto actual training image patches, explaining decisions through similarity to concrete visual prototypes ("this looks like that") without manual labels.

2. **Unsupervised Concept Basis Models (Self-Explaining Neural Networks – SENN):** Decomposes predictions into an unsupervised concept basis $h(x)$ and locally linear relevance coefficients $\theta(x)$, regularized by an autoencoder reconstruction objective and Lipschitz smoothness constraints.
3. **Foundation Model / VLM-Assisted Concept Discovery (Label-Free CBMs):** Leverages Large Language Models to generate concept vocabularies and Vision-Language Models (CLIP) to perform zero-shot concept scoring, creating pseudo-labels to train a Concept Bottleneck Model.
4. **Unsupervised Latent Clustering and Sparse Dictionary Learning (ACE / SAE):** Clusters intermediate feature activations of segmented image patches to identify recurring visual concepts automatically.

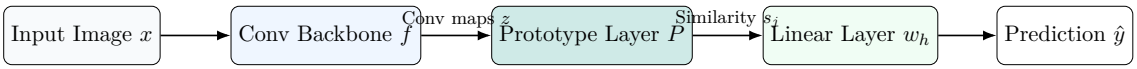
✍ **Comprehensive Technical Walkthrough**

1. Challenge Context

Standard Concept Bottleneck Models (CBMs) enforce an architectural bottleneck $x \xrightarrow{g} c \xrightarrow{f} y$. However, training the concept encoder $g(x)$ conventionally demands costly, exhaustive ground-truth concept annotations $c^* \in \{0,1\}^K$ for every sample. In the absence of concept annotations, an explainable-by-design architecture must discover or synthesize concepts autonomously.

2. Approach 1: Prototype-Based Learning (ProtoPNet — "This Looks Like That")

Chen et al. (2019) introduced Prototype Part Networks (ProtoPNet), an inherently interpretable architecture that substitutes abstract symbolic concepts with **case-based visual prototypes**:



- **Mechanism:** The network learns a set of M prototype vectors $\{p_j\}_{j=1}^{M}$ in the convolutional latent space. For an input image patch z, the prototype layer calculates an inverted L_2 distance similarity:

$$s_j(z) = g(d(z, p_j)) = \log\left(\frac{\|z - p_j\|_2^2 + 1}{\|z - p_j\|_2^2 + \epsilon}\right) \tag{12}$$

  The output is a strictly additive linear combination: $\hat{y} = \sum_{j=1}^{M} w_j \max_z s_j(z)$.

- **Push Projection Step:** During training, every learned latent prototype vector p_j is periodically *pushed* (projected) to the exact latent representation of the nearest training image patch:

$$p_j \leftarrow \underset{z \in \text{patches}(\mathcal{D})}{\arg\min} \|z - p_j\|_2 \tag{13}$$

- **Interpretability Without Labels:** The prototypes act as concrete, self-contained visual concepts (e.g., *"red throat of a ruby-throated hummingbird"*). The model explains decisions by displaying the exact training patches that match the input regions, requiring zero human concept labels during training.

3. Approach 2: Unsupervised Concept Basis (Self-Explaining Neural Networks — SENN)

Alvarez-Melis & Jaakkola (2018) proposed SENN, an architecture that enforces a generalized linear model formulation over learned basis concepts:

$$f(x) = \sum_{j=1}^{K} \theta_j(x) h_j(x) \tag{14}$$

- **Concept Encoder ($h : \mathcal{X} \to \mathbb{R}^K$):** Maps raw input x to K basis concepts without supervision. To ensure that $h(x)$ captures meaningful data factors, SENN incorporates an **autoencoder reconstruction loss**:

$$\mathcal{L}_{\text{rec}} = \|x - \text{Decoder}(h(x))\|_2^2 \tag{15}$$

- **Parameterizer Network ($\theta : \mathcal{X} \to \mathbb{R}^K$):** Generates instance-specific relevance scores. To guarantee interpretability, $\theta(x)$ is regularized via a **robustness penalty** enforcing local Lipschitz continuity:

$$\mathcal{L}_{\text{robust}} = \|\nabla_x f(x) - \theta(x)\|_2^2 \tag{16}$$

  This ensures that the relevance weights $\theta_j(x)$ remain stable under small input perturbations, preventing adversarial explanation instability.

- **Limitation:** Unsupervised basis concepts $h_j(x)$ correspond to mathematical principal directions that do not always align cleanly with human-nameable semantic concepts.

4. Approach 3: Foundation Model / Vision-Language Model Concept Synthesis

Recent advances (Oikarinen et al., 2023; Yuksekgonul et al., 2023) utilize multimodal foundation models to eliminate the concept annotation bottleneck:

Step 1. LLM Concept Vocabulary Discovery: Query a generative Large Language Model (e.g., GPT-4) with prompt templates: *"List 30 visual, distinct anatomical attributes useful for identifying thoracic diseases on X-rays."* The LLM outputs a structured semantic dictionary $\mathcal{C} = \{c_1, \ldots, c_K\}$.

Step 2. VLM Zero-Shot Pseudo-Labeling: Pass each unlabeled training image x through a Vision-Language Model (e.g., CLIP, BiomedCLIP). Compute the cosine similarity between the image embedding and text concept prompt embeddings:

$$\tilde{c}_j(x) = \frac{\langle E_{\text{vision}}(x), E_{\text{text}}(\text{“a photo showing } c_j\text{”}) \rangle}{\|E_{\text{vision}}(x)\| \|E_{\text{text}}\|} \tag{17}$$

Step 3. Train a Label-Free CBM: Train an explainable-by-design Concept Bottleneck Model $x \xrightarrow{g} c \xrightarrow{f} y$ using these synthesized continuous pseudo-labels $\tilde{c}$, achieving a faithful concept bottleneck without a single human-annotated concept label.

5. Comparative Matrix of Label-Free Concept Paradigms

Paradigm	Concept Nature	Supervision Required	Interpretability	Intervention
ProtoPNet	Case-based visual image patches	Class labels only	High (visual exemplar comparison)	Prototype pruning / patch swapping
SENN	Continuous latent basis vectors	Class labels only + autoencoder	Moderate (latent factor semantic gap)	Direct numerical basis manipulation
VLM-CBM	Human-nameable text concepts	Class labels + frozen foundation model	High (standard clinical/semantic lexicon)	Full test-time concept intervention

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

In official grading, students lost a full point (scored 2.00 / 3.00) because:

- They lumped unsupervised techniques into a single vague phrase without distinguishing between the two distinct foundational families: **prototype-based** (ProtoPNet) versus **unsupervised concept basis** (SENN).
- They omitted the specific architectural mechanisms (e.g., ProtoPNet push projection, SENN reconstruction loss, VLM zero-shot concept similarity). Both families must be detailed explicitly.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 7: "Concept-based Explainable AI", Section: "Explainable-by-Design Concept-based Models: ProtoPNet and SENN" (pp. 100–108).

¶ Alternative Perspectives & Methodological Variants

Automated Concept-based Explanations (ACE) (Ghorbani et al.) segments images across multiple resolutions, extracts intermediate layer activations, clusters them via k-means, and filters out non-salient clusters using TCAV scores, automatically discovering concepts without prior annotations.

Question 6: Adversarial Attacks in Trustworthy AI: Threat Feasibility and Real-World Impact

Open Inquiry | 2.50 Points

Why are adversarial attacks important in trustworthy AI? Are they a real threat to AI models? Motivate your answer with at least two examples or reasons.

✓ Executive Solution Summary

Adversarial attacks are a cornerstone of Trustworthy AI research because they expose the fundamental divergence between statistical pattern matching and human cognitive reasoning, proving that high empirical validation accuracy does not equate to robustness or safety.
Yes, adversarial attacks are an undeniable, demonstrated real-world threat that extends far beyond digital pixel manipulations. Utilizing Expectation Over Transformation

(EOT), attackers manufacture physical-world perturbations that survive variable lighting, camera angles, and print artifacts. Key real-world demonstrations include: (1) physical stickers fooling autonomous vehicle traffic sign recognition, (2) adversarial patterns on clothing evading automated surveillance weapon detectors, (3) dermatological image perturbations enabling healthcare insurance fraud, and (4) universal token jailbreaks bypassing safety guardrails in Large Language Models.

✍ **Comprehensive Technical Walkthrough**

1. Theoretical Significance in Trustworthy AI

Adversarial attacks fundamentally challenge the foundations of deep learning:

- **Failure of Empirical Risk Minimization (ERM):** Standard training minimizes average loss over independent and identically distributed (i.i.d.) data: $\min_\theta \frac{1}{N}\sum_{i=1}^{N}\mathcal{L}(f_\theta(x_i), y_i)$. Adversarial attacks prove that models minimize loss by exploiting non-robust, high-frequency statistical shortcuts that vanish under adversarial worst-case evaluation.
- **The Security Arms Race:** Most heuristic defenses (such as gradient masking or defensive distillation) provide an illusion of security that is quickly bypassed by adaptive attacks (e.g., Carlini-Wagner, AutoAttack). This demonstrates that safety cannot be achieved by ad-hoc patches, but demands certified, mathematically robust formulations.
- **Regulatory and Societal Trust:** Under the European AI Act (Regulation 2024/1689, Article 15), high-risk AI deployments are legally mandated to be resilient against adversarial attempts to manipulate model outputs.

2. Are Adversarial Attacks a Real Threat? (Physical Feasibility)

Skeptics originally argued that adversarial perturbations were mathematical artifacts restricted to digital 8-bit image matrices that would dissolve when photographed by a real-world camera. This objection was decisively refuted by **Expectation Over Transformation (EOT)** (Athalye et al., 2018), which optimizes perturbations across a distribution of physical transformations $\mathcal{T}$:

$$\max_{\delta} \mathbb{E}_{t\sim\mathcal{T}}\left[\mathcal{L}\left(f_\theta(t(x+\delta)), y\right)\right] \quad \text{s.t.} \quad \|\delta\|_p \leq \epsilon \tag{18}$$

where $\mathcal{T}$ models viewing angles, ambient lighting variations, distance scaling, motion blur, and print color distortions. Perturbations optimized under EOT survive real-world physical rendering.

3. Exhaustive Real-World Threat Scenarios

Domain	Threat Mechanism	Demonstrated Physical Exploit	Catastrophic Consequence
Autonomous Driving	Physical Adversarial Patches & Stickers	A standard STOP sign is affixed with small, strategically placed stickers. Computer vision models (YOLO / MobileNet) classify it as a "Speed Limit 80 km/h" sign.	Autonomous vehicle enters an intersection at full speed without braking, risking fatal collisions.
Surveillance & Security	Adversarial Camouflage Clothing (Adv-Shirt)	Fabric printed with adversarial patterns optimized to suppress bounding-box proposals in Faster R-CNN and YOLO.	Armed intruder wearing the garment becomes completely invisible to automated airport surveillance systems.
Healthcare & Insurance	Imperceptible High-Frequency Perturbations	Medical billing fraud: an imperceptible perturbation added to a benign dermoscopy skin lesion flips the AI diagnostic classifier to "Malignant Melanoma".	Fraudulent insurance payouts or unnecessary invasive surgical procedures.
Generative AI / LLMs	Universal Adversarial Suffixes (GCG Attack)	Automated token sequences (e.g., Zou et al.) appended to prompts bypass alignment, RLHF, and system guardrails.	LLM produces actionable instructions for synthesizing dangerous pathogens or cyber-warfare exploits.

4. Transferability: Amplifying the Threat Vector

The danger of adversarial attacks is magnified exponentially by **cross-model transferability**: perturbations generated against an accessible, open-source surrogate architecture (e.g., ResNet-50) consistently fool proprietary, closed-source commercial black-box models trained on completely different datasets. An attacker does not need access to internal weights or APIs to execute a catastrophic attack.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

To achieve full marks (2.50 / 2.50), students must emphasize that adversarial attacks are NOT merely synthetic digital bugs, but real physical threats. Mentioning physical perturbations on road signs, weapon detection evasion via printed textiles, and the absence of universal defenses ensures maximum score.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 12: "Adversarial Attacks and Defenses", Sections: "Why are Adversarial Attacks Important?", "Is the Threat Real?", and "Transferability: Not Network Specific" (pp. 175–182).

¶ Alternative Perspectives & Methodological Variants

Universal Adversarial Perturbations (UAPs) (Moosavi-Dezfooli et al.) generate a single, image-agnostic perturbation vector $\mathbf{v}$ such that $f(x + \mathbf{v}) \neq f(x)$ for over 80% of all natural images, demonstrating that vulnerability is an intrinsic geometric property of deep neural

representations.

Question 7: Credit Card Fraud Detection on Tabular Data: XAI Methods and System Recommendation Open Inquiry 5.00 Points

A financial institution uses an AI system to detect potentially fraudulent credit card transactions. The model operates on tabular data, such as transaction amount, merchant category, location, time of day, customer history, and recent spending patterns. Since incorrect predictions may block legitimate transactions or allow fraud to pass undetected, the institution wants the system to provide understandable explanations to fraud analysts.
Describe at least three explainability methods that could be used to help analysts understand the decisions of the system. Discuss the types of explanation representation that would be most appropriate for this scenario (e.g., feature importance, local rules). Choose one explainability method you would recommend for this use case and clearly justify your choice based on the needs of fraud analysts.

✓ **Executive Solution Summary**

Financial fraud detection on tabular data requires post-hoc, local explainability methods capable of providing rapid, actionable, and legally defensible explanations under strict latency constraints. We evaluate three distinct methods: (1) **SHAP** (TreeSHAP/KernelSHAP for axiomatic feature attribution), (2) **LORE** (Local Rule-Based Explanations via genetic neighborhood generation and local decision trees), and (3) **Counterfactual Explanations** (Wachter et al. / DiCE for actionable recourse).
We evaluate three explanation representations: feature importance, local decision rules, and counterfactual rules/examples. We formally **recommend LORE** (optionally paired with TreeSHAP), because its dual output of **symbolic IF-THEN local decision rules** and **counterfactual rules** perfectly aligns with the cognitive operational workflow of fraud analysts and satisfies regulatory mandates for adverse action notices under financial consumer protection laws.

✍ **Comprehensive Technical Walkthrough**

1. Operational Context & Tabular Data Characteristics

Credit card fraud detection systems operate on heterogeneous tabular features:

- *Continuous numerical features:* Transaction amount, velocity of transactions in the last 1 hour, geographical distance from last transaction.
- *High-cardinality categorical features:* Merchant Category Code (MCC), country code, device fingerprint.
- *Temporal and historical aggregations:* Deviation from 30-day average spending, hour of day.

Stakeholder Profile: The Fraud Analyst

Fraud analysts operate under extreme operational pressure: they review hundreds of flagged transactions daily, spending on average **30 to 60 seconds per case**. They require explanations that:

a. Provide instant cognitive clarity without requiring statistical parsing.

b. Supply an actionable verification checklist when contacting the cardholder.

c. Generate legally compliant audit trails (satisfying GDPR Article 22 and the Equal Credit Opportunity Act for adverse action notices).

2. In-Depth Description of Three Explainability Methods

Method 1: SHAP (SHapley Additive exPlanations — TreeSHAP)

- **Mechanism:** SHAP models local explanations via an additive linear surrogate $g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z'_j$, where each feature attribution ϕ_j equals the exact Shapley value computed across feature coalitions. In financial tree ensembles (e.g., XGBoost, LightGBM, CatBoost), **TreeSHAP** computes exact values in polynomial time $\mathcal{O}(TLD^2)$.
- **Representation:** Feature attribution bar/waterfall plot showing signed contributions: e.g., $\phi_{\text{Amount}} = +0.35$, $\phi_{\text{ForeignMCC}} = +0.28$, $\phi_{\text{ChipUsed}} = -0.15$.
- **Limitation in Fraud Operations:** Feature attribution rankings tell the analyst which features drove the score, but **do not provide explicit operational thresholds or logical condition combinations**.

Method 2: LORE (Local Rule-Based Explanations)

- **Mechanism:** Guidotti et al. designed LORE specifically for tabular black-box models:
  1. *Local Neighborhood Generation:* Uses an active learning **genetic algorithm** to generate a dense, balanced synthetic neighborhood Z of instances around the target transaction x, maintaining empirical feature correlation and sampling across both sides of the local decision boundary.
  2. *Local Surrogate Fitting:* Trains an interpretable surrogate **Decision Tree** c on Z, labeled by the black-box model $f(Z)$.
  3. *Rule Extraction:* Traverses the path from the root to the leaf node corresponding to x to extract two complementary symbolic structures:
     - **Local Decision Rule ($r = p \rightarrow \hat{y}$):** A conjunction of premises explaining why the transaction was flagged:

$$\begin{aligned} r : (\text{Amount} > \$1500) \wedge (\text{Country} \neq \text{HomeCountry}) \\ \wedge\, (\text{CardPresent} = \text{False}) \implies \text{Fraud} \end{aligned} \tag{19}$$

     - **Set of Counterfactual Rules ($\delta = p' \rightarrow y'$):** Minimal condition modifications that flip the prediction back to legitimate:

$$\delta : (\text{TwoFactorAuth} = \text{Verified}) \vee (\text{Merchant} = \text{Whitelisted}) \implies \text{Legitimate} \tag{20}$$

- **Strength:** Captures non-linear thresholds and feature conjunctions natively without assuming linearity.

Method 3: Counterfactual Explanations (Wachter et al. / DiCE)

- **Mechanism:** Finds the closest realistic tabular profile x^* that reverses the fraud classification to legitimate, solving:

$$x^* = \arg\min_{x'} d_{\text{MAD}}(x, x') + \lambda \left(f(x') - y_{\text{legit}}\right)^2 \quad \text{s.t.} \quad x'_j = x_j \; \forall j \in \mathcal{X}_{\text{immutable}} \tag{21}$$

where distance is normalized by the Median Absolute Deviation (d_{MAD}) to handle mixed continuous and categorical tabular scales.

- **Representation:** Example-based tabular row comparison showing: *"If Amount had been* $\leq \$450$ *instead of* $\$1800$*, the transaction would have passed without intervention."*

3. Comparative Evaluation of Explanation Representations

Representation	Cognitive Format	Pros for Fraud Analysts	Cons for Fraud Analysts
Feature Importance	Ranked bar charts of signed weights (ϕ_j)	Instant high-level visual triage; identifies dominant factors	Lacks operational thresholds; does not state when a flag clears
Local Decision Rules	Symbolic IF-THEN logic: $\bigwedge_i (x_i \in [a, b]) \rightarrow \hat{y}$	Perfectly mirrors fraud policy rules; intuitive, deterministic, and auditable	Tree depth must be constrained to prevent cognitive overload
Counterfactual Rules	Minimal premise delta: $p \rightarrow p'$	Direct operational verification script for analyst-customer calls	May suggest unrealistic actions if manifold constraints are weak

4. Final Recommendation and Operational Justification

Recommendation: We formally recommend deploying **LORE (Local Rule-Based Explanations)** as the primary explainer, complemented by TreeSHAP summary waterfalls on the analyst dashboard.

Justification Based on Fraud Analyst Operational Requirements:

J1. Cognitive Alignment with Standard Operating Procedures (SOP): Fraud analysts are trained on rule engines (e.g., Falcon rules). A local decision rule (r) provides a structured, logical proposition that can be evaluated in under 10 seconds:

$$\textbf{Rule}: \quad [\text{Amount} > \$1200] \wedge [\text{Time} \in [02:00, 05:00]] \wedge [\Delta_{\text{Location}} > 500 \text{ km}] \implies \textbf{Fraud}$$

This eliminates cognitive guesswork regarding feature interactions.

J2. Actionable Verification Script via Counterfactual Rules (δ): When an analyst phones a cardholder to verify a suspicious transaction, the counterfactual rule provides the exact verification questions needed: *"Did you authorize an overseas online purchase between 2 AM and 5 AM?"* If the user confirms secondary authentication, the transaction is safely released.

J3. Handling Non-Linear Step Functions in Tabular Data: Credit card fraud boundaries are dominated by hard thresholds (e.g., credit limits, regulatory reporting thresholds at $\$10,000$). Linear surrogate models (such as LIME) smooth out these step boundaries, whereas LORE's local decision tree models them with exact fidelity.

J4. Regulatory Compliance for Adverse Action Notices: Under Article 22 of the GDPR and Fair Lending Regulations, financial institutions blocking credit or accounts must provide human-understandable, specific reasons. LORE's symbolic decision rules directly constitute legally valid adverse action statements.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

This answer earned a perfect score (5.00 / 5.00) in previous sessions. Key scoring factors include:

- Explicitly describing at least three distinct methods with mathematical/algorithmic mechanisms.
- Thoroughly addressing tabular-specific challenges (categorical variables, thresholds, nonlinearities).
- Clearly distinguishing between local decision rules and counterfactual rules when presenting LORE.
- Structuring the justification around the operational realities of fraud analysts (30-second triage, customer phone verification, legal compliance).

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 5: "Local Surrogate Models: LORE" (pp. 45–47), "SHAP: TreeSHAP" (pp. 58–60), and Chapter 9: "Counterfactual Explanations" (pp. 130–138).

¶ Alternative Perspectives & Methodological Variants

Anchors (Ribeiro et al., 2018) provides an alternative rule-based approach by computing high-precision IF-THEN rules that "anchor" the prediction such that changes to other features do not alter the outcome with high statistical probability $(1 - \delta)$.